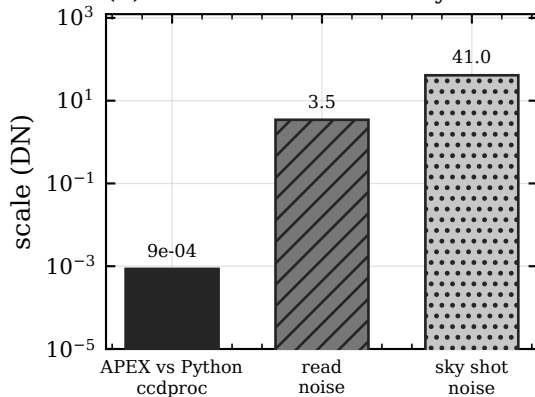


(a) stage-wise pixel audit

stage	max $ \Delta $ (DN)	robust σ (DN)	result
master bias	0	0	bit exact
master dark	0	0	bit exact
master flat	0	0	bit exact
bias subtract	0	0	bit exact
dark subtract	0	0	bit exact
flat field	0	0	bit exact
full pipeline	8.61e-04	2.58e-05	float32 round

master construction + bias/dark/flat application; zero means every pixel matched

(b) below detector and sky noise



NGC 6811 *B* single 60 s · Moravian C3-61000 (2×2) · night 2026-06-11 · 8 bias / 8 dark (60 s) / 5 flat

APEX vs astropy ccdproc 2.5.1 · cosmetic step off (validated by injection) · generator: calib_crosscheck_ngc6811.py